

A Public Benchmark Dataset for TCR-pMHC Prediction

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Abstract

Accurate prediction of T cell receptor (TCR) specificity for cognate peptides presented by the major histocompatibility complex (pMHC) is a key challenge in understanding adaptive immunity and enabling immune-based therapeutics and diagnostics. Successful models predicting TCR-pMHC specificity must be able to generalize to novel TCRs and novel pMHCs not observed during model training. Here, we provide a labeled dataset intended for general use as a test of TCR-pMHC model generalization to novel TCRs specific for novel pMHCs.

Introduction

Predicting the specificity of a TCR to its cognate pMHC is a central problem in immunology. Accurate models capable of making predictions for novel TCRs and pMHCs not observed over the course of training would be a valuable advance in our understanding of adaptive immunity. Such models would circumvent the need for expensive, time-consuming, and low-throughput experimental methods and pave the way to enabling new therapeutic and diagnostic modalities.

Many models have been proposed to predict this interaction. However, there is not a consistent benchmark available for testing these models' ability to generalize. The 2025 IMMREP competition [1] offered a direct comparison of many models on this important problem. Here, we provide the full labeled evaluation dataset used in that competition under a permissive MIT License. This dataset can serve now as a common point of comparison for new and existing models of the TCR-pMHC interaction.

Dataset Composition

This dataset comprises a total of 10,000 records divided between 1,000 positive labels and 9,000 negative labels. The positive labels comprise 1,000 unique TCRs each specific to one of 20 distinct peptide-MHC targets (50 TCRs per target). These peptides were drawn from 2 MHC context, with 10 peptides coming from each of HLA-A*02:01 and HLA-B*40:01. The peptides themselves are all 9-mer epitopes from a variety of viral sources (Table 1). These peptides are all novel relative to those with previously published and collated T cell epitope datasets. Specifically, this dataset includes only peptides with little similarity to previously published epitopes as found in IEDB [2], VDJdb [3], McPAS [4], or previously published MIRA data [5]. All peptides are 4 or more Levenshtein distance from any published peptide and share no substring longer than 5 consecutive amino acids with any published peptide.

For each MHC, the dataset includes 50 TCRs which were experimentally determined to be specific for that pMHC by MIRA [6] and pairSEQ [7] (see Data Generation below for more details). Each of these pMHC-specific TCRs was determined to not be specific to any of the other 19 pMHCs in the dataset (as described in Data Generation). The complete labeled dataset consists of these 1,000 positive TCR-pMHC combinations and the 9,000 negative TCR-pMHC combinations formed by matching each TCR to all alternative peptides within its ground truth MHC context.

Data Generation

The positive and negative labels in this dataset correspond to TCR-pMHC combinations observed to result in T cell activation (positive labels) and to not result in T cell activation (negative labels). The data was generated as follows. PBMCs from HLA-genotyped donor leukopaks were isolated via negative selection for naïve, CD8+ T cells, and incubated with a pool of peptides for T cell expansion. A portion of this expanded T cell population was restimulated with the entire pool of peptides and input into a pairSEQ experiment to determine TCRA-TCRB pairings as described in Howie et al [7]. The remainder of the peptide pool expanded T cell sample was input into a MIRA (Multiplex Identification of T cell Receptor-Antigen specificity) experiment from which we determined TCRB-peptide specificities as described in Klinger et al [6], Nolan et al [5] (experimental methods) and Snyder et al (statistical methods) [8].

From the TCRA-TCRB pairings (from pairSEQ) and TCRB-peptide specificities (MIRA), we constructed complete TCR-peptide information. Finally, we defined the MHC context using peptide-MHC binding prediction from NetMHCpan4.1 [9] and limiting to data in which the purported presenting MHC for the peptide was also present in the source donor's HLA genotype. In combination, this process generated the positive-label TCR-pMHC data.

For the negative-label TCR-pMHC data, we took advantage of the multiplexed nature of the MIRA assay (figure 1). In MIRA, TCRB clonotypes are assessed for activation against all peptide pools in the experiment and only reported as positive for a given peptide when their set of activated pools corresponds to the unique occupancy pattern of that peptide. Consequently, any TCRB reported as positive must not activate in response to any other peptide in the experiment, as this would result in activation in additional pools and disqualify it from being labeled as a positive. Since all 20 peptides in the evaluation dataset were assessed simultaneously in each MIRA experiment, every positive TCR is a valid negative for the remaining peptides.

Dataset Usage

The dataset is provided for general use under an MIT License and can be downloaded from DOI: 10.6084/m9.figshare.30491582. The schema is described in Table 2.

References

- [1] IMMREP25 Organizers. Immrep25: Tcr specificity prediction challenge. <https://kaggle.com/competitions/immrep25>, 2025. Kaggle. (IMMREP)
- [2] <https://academic.oup.com/nar/article/53/D1/D436/7903371> (IEDB)
- [3] <https://academic.oup.com/nar/article/46/D1/D419/4101254> (VDJdb)
- [4] <https://academic.oup.com/bioinformatics/article/33/18/2924/3803440> (McPAS)
- [5] <https://pmc.ncbi.nlm.nih.gov/articles/PMC7418738/> (Nolan MIRA)
- [6] <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0141561> (Klinger MIRA)
- [7] <https://pubmed.ncbi.nlm.nih.gov/26290413/> (Howie pairSEQ)
- [8] <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2024.1488860/full> (Snyder MIRA)
- [9] <https://academic.oup.com/nar/article/48/W1/W449/5837056> (NetMHC)

Peptide	HLA	Taxon	Min Levenshtein Distance	Max Shared Substring
FLDCKSYIL	A*02:01	Human respirovirus 1	4	3
HLDDYPYLM	A*02:01	Human betaherpesvirus 5	5	3
ILHTHVPEV	A*02:01	Human alphaherpesvirus 3	5	4
ILMHATYFL	A*02:01	Human alphaherpesvirus 1	4	3
KLYPFLWFA	A*02:01	Human betaherpesvirus 6b	5	3
MTDYDYLEV	A*02:01	Human betaherpesvirus 5	5	4
SQFNWTIYL	A*02:01	Human betaherpesvirus 6b	4	3
TVYPYGTSL	A*02:01	Human betaherpesvirus 5	5	3
YLFNADIWI	A*02:01	Human betaherpesvirus 6b	5	3
YMFYDGYDV	A*02:01	Rhinovirus c	5	3
FEAQPGALL	B*40:01	Human betaherpesvirus 5	5	4
GENALTYAL	B*40:01	Human alphaherpesvirus 1	5	3
GEVLACYAL	B*40:01	Human alphaherpesvirus 3	4	3
MEMPDYLLL	B*40:01	Rotavirus a	5	4
MENWSALEL	B*40:01	Human betaherpesvirus 5	5	4
REDDYSWWL	B*40:01	Human betaherpesvirus 5	4	4
SEESAFYVL	B*40:01	Human parvovirus b19	5	4
YEDGVIFYL	B*40:01	Human betaherpesvirus 6b	5	4
YENGSTPVL	B*40:01	Human betaherpesvirus 5	5	3
YESYIPGAL	B*40:01	Human betaherpesvirus 5	4	3

Table 1: The 20 peptides represented in this dataset, described by their HLA context, source taxon, minimum Levenshtein distance to any published epitope, and maximum shared substring with any published epitope.

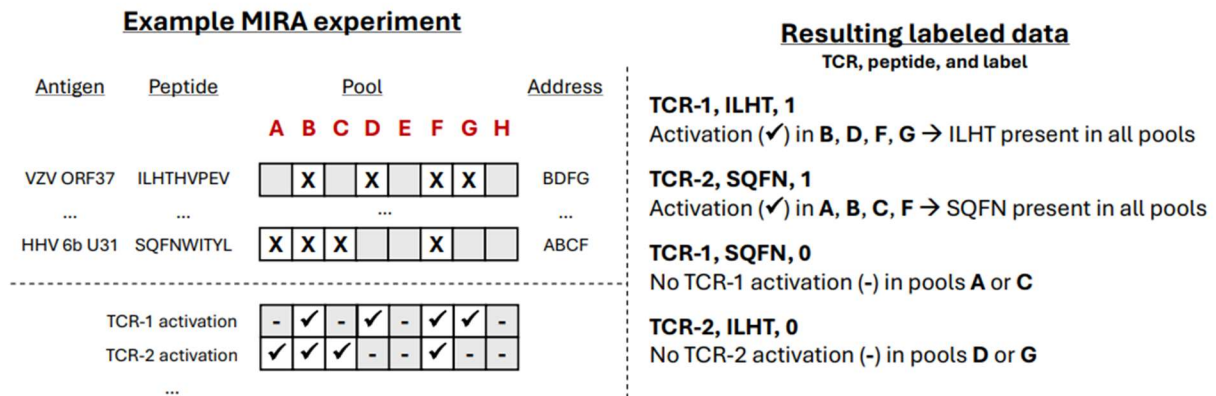


Figure 1: Negative label generation. A MIRA experiment consists of a set of pools. Each peptide in the experiment is assigned to a unique subset of these pools, termed its “Address”. Each TCR in the reaction is then assessed for activation in each of the pools. A TCR is concluded to be specific for a peptide if and only if its activation pattern across the pools is concurrent with the peptide uniquely assigned to those pools. Consequently, a TCR called as specific for one peptide must necessarily not activate against the other peptides in the experiment, else activation would also be observed in pools where the called peptide is not present.

Column	Type	Description
tcra_bioidentity	string	TCR alpha represented as <CDR3 amino acid sequence>+<V gene>+<J gene>
tcrb_bioidentity	string	TCR beta represented as <CDR3 amino acid sequence>+<V gene>+<J gene>
tcra_cdr1	string	CDR1 alpha amino acid sequence
tcra_cdr2	string	CDR2 alpha amino acid sequence
tcra_cdr3	string	CDR3 alpha amino acid sequence
tcrb_cdr1	string	CDR1 beta amino acid sequence
tcrb_cdr2	string	CDR2 beta amino acid sequence
tcrb_cdr3	string	CDR3 beta amino acid sequence
tcra_v	string	TCR alpha V gene
tcra_j	string	TCR alpha J gene
tcrb_v	string	TCR beta V gene
tcrb_j	string	TCR beta J gene
tcra_trimmed	string	Amino acid sequence of full variable domain of TCR alpha
tcrb_trimmed	string	Amino acid sequence of full variable domain of TCR beta
peptide	string	Peptide Amino acid sequence
hla	string	HLA name (e.g. "A*02:01")
hla_sequence	string	Full HLA amino acid sequence
hla_trimmed	string	Amino acid sequence of the peptide binding domain
label	integer	1 (positive) or 0 (negative)

Table 2: Dataset schema.